Efficient Real Style Transfer: Latent Bridge Matching via Optimal Transport

Anonymous submission

Abstract

Art-to-art style transfer is susceptible to overstated claims: because the source image is already artwork, the unchanged image can score highly for many target styles, creating a trivial failure mode invisible to raw style affinity. We expose this failure mode through identical-image transfer (IDT) calibration, which measures style gain only above the unchanged-image floor. Under this protocol, we introduce *Latent Bridge Matching* (LBM), an efficient latent transport family that achieves real style transfer at consumer cost. LBM learns the minimum-energy endpoint shift that exceeds the IDT floor, regularized by kinetic control and terminal distribution matching to preserve structure. On Distinct5-WikiArt—a deliberately hard five-style subset with the lowest IDT CLIP-Style score—LBM with kinetic regularization exceeds the calibrated floor after 1.2 minutes on a single RTX 3060, while reproduced baselines remain below it even after multi-hour training. LBM with balanced regularization achieves the strongest balanced operating point, and inference-time style controls further shift a fixed checkpoint without retraining, indicating that part of the remaining gap is attributable to execution rather than representation. These results demonstrate that LBM provides an efficient pathway to real style transfer, combining target-aligned motion, minute-scale training, and practical operating points on consumer hardware.

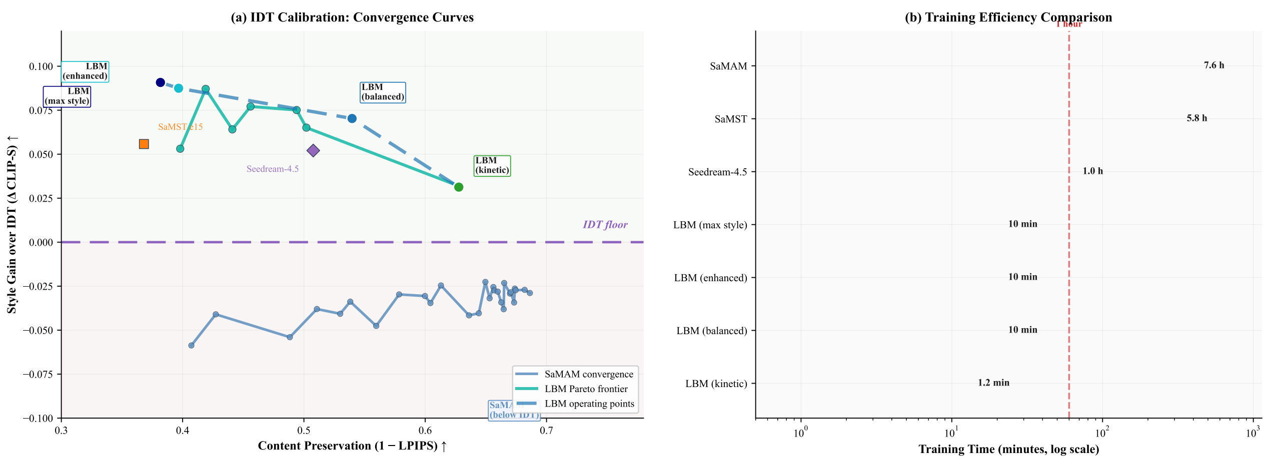

Figure 1: Real style transfer requires both target-aligned motion and low computational cost. Panel (a) shows the transfer plane: SaMAM remains below the IDT floor, failing to shift reliably toward the requested style, while the LBM family and inference-time style controls occupy the positive target-aligned region above zero. Panel (b) compares training time on a logarithmic scale: LBM configurations achieve minute-scale training on a single RTX 3060, while reproduced baselines require multi-hour training. The vertical dashed line marks the one-hour boundary.

Introduction

Domain-level artistic transfer requires a target *style identity*, not a target exemplar. We study the most constrained practical version: given only a source artwork and a target style identifier, a compact model must produce *real* style transfer—a shift in the correct target-style direction—without per-image optimization or a large pretrained generator. This task is susceptible to overstated claims: because the source image is already artwork, the unchanged image can score highly for many target styles. We expose this failure mode through identical-image transfer (IDT) calibration, which measures style gain only above the unchanged-image floor.

We introduce *Latent Bridge Matching* (LBM), a compact VAE-space transport family in which endpoint pairing determines *where* to shift, a style-conditioned latent field learns *how* to shift, and terminal distribution matching plus kinetic control regularize that shift. The design is supported by four theoretical results: kinetic regularization selects the minimum-energy endpoint, Euler integration maintains first-order accuracy, optimal batch pairing minimizes endpoint displacement variance, and semantic-aligned sliced Wasserstein distance enforces binwise distributional consistency. On Distinct5-WikiArt, LBM exceeds the calibrated floor after 1.2 minutes on a single RTX 3060, while SaMAM remains below it after 7.6 hours. Contributions: (1) IDT-calibrated evaluation protocol showing raw style affinity can overstate control; (2) LBM, a style-ID-only transport family achieving real transfer under minute-scale training; (3) inference-time style controls that improve a fixed checkpoint without retraining.

Related Work

Prior style transfer methods fall into exemplar-guided stylization, which consumes a target image at test time (Gatys, Ecker, and Bethge 2016; Huang and Belongie 2017; Deng et al. 2022; Hong et al. 2023; Huang et al. 2024), and large-prior editing, which inherits a strong pretrained generator (Zhang et al. 2024; Chung, Hyun, and Heo 2024). Our setting is closer to compact multi-style transfer (Liu et al. 2024b; Botti et al. 2025; Liu et al. 2025), where inference receives only a source image plus a target style identity. Technically, LBM draws on flow matching (Lipman et al. 2023; Liu et al. 2024a) and optimal transport (Arjovsky, Chintala, and Bottou 2017; Tong et al. 2024; Pooladian and Niles-Weed 2023; Bernoully, Lavenant, and Staerman 2024; Chen et al. 2022), but uses transport as supervisory geometry rather than as a claim of online OT solving (Lu et al. 2026). Because unchanged artworks can score highly for many target styles, raw style affinity is insufficient (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022; Yang et al. 2022b); we evaluate style gain only above IDT.

Method

Training objective and inference contract

LBM is designed around two constraints that prior work often relaxes: strict style-ID-only inference and minute-scale efficient training. The model operates in Variational Autoencoder (VAE) latent space and receives only a source image plus a target style identifier at test time. Figure 2 separates this simple interface from the training-side machinery used to achieve it. For Distinct5-WikiArt, the source latent is $z_0 \in \mathbb{R}^{4\times 64\times 64}$; target-domain latents appear only as supervision during training.

The method follows one principle: learn the most efficient endpoint shift that exceeds the IDT floor, then regularize that shift to preserve structure. Transport serves as supervisory geometry rather than as a claim of online optimal transport solving (Lu et al. 2026). In the Distinct5 configuration, LBM variants use endpoint-mode flow matching:

$$\hat{v} = v_\theta(z_0, 1, s), \qquad \hat{z}_1 = z_0 + \hat{v}, \tag{1}$$

and optimize the endpoint objective

$$\mathcal{L}_{\text{active}} = \lambda_{\text{term}}\mathcal{L}_{\text{term}}(\hat{z}_1) + \lambda_{\text{kin}}\|\hat{v}\|_2^2, \tag{2}$$

where the precomputed endpoint set supplies terminal targets and the parent flow residual is disabled. This formulation keeps the active path efficient: one endpoint prediction, one terminal distributional objective, and explicit kinetic control.

Design-grounding theory. The next four results formalize the design choices that make LBM both efficient and well-behaved: minimum-energy endpoint motion, stable explicit integration, low-cost endpoint pairing, and binwise terminal distribution consistency.

Theorem 1 (Minimum-energy endpoint selection)**.** *Let $\mathcal{S}_s = \{z : \mathcal{L}_{\text{term}}(z, s) = 0\}$ denote the zero-terminal-loss set for target style s. Under endpoint-mode flow matching, any global minimizer of Eq.* (2) *over $\mathcal{S}_s$ is $z_1^\star = \arg\min_{z\in\mathcal{S}_s} \|z - z_0\|_2^2$. LBM selects the smallest displacement from the source latent.*

Theorem 2 (Euler stability of the active path)**.** *Assume $v_\theta(\cdot, t, s)$ is L-Lipschitz in z and continuously differentiable in t. Let $z(t)$ solve $\dot{z} = v_\theta(z, t, s)$ with $z(0) = z_0$, and let the Euler iterates satisfy $h_{k+1} = h_k + \Delta v_\theta(h_k, t_k, s)$ with $\Delta = 1/K$. Then there exists a constant C, independent of K, such that* $\max_{0\le k\le K} \|h_k - z(t_k)\|_2 \le C/K$.

Theorem 3 (Optimal batch pairing lowers target second moment)**.** *Consider a batch of source latents $\{z_0^{(i)}\}_{i=1}^B$ and candidate target latents $\{z_1^{(j)}\}_{j=1}^B$. Let $\pi^\star = \arg\min_{\pi\in S_B}\sum_{i=1}^B \|z_1^{(\pi(i))} - z_0^{(i)}\|_2^2$. Then $\frac{1}{B}\sum_{i=1}^B \|z_1^{(\pi^\star(i))} - z_0^{(i)}\|_2^2 \le \frac{1}{B}\sum_{i=1}^B \|z_1^{(\pi(i))} - z_0^{(i)}\|_2^2$ for every other bijection π. The optimal transport assignment minimizes the batchwise second moment of endpoint targets.*

Interpretation. Theorems 1–4 explain why LBM is both efficient and well-behaved. Theorem 1 shows that kinetic regularization is not a heuristic penalty: once the terminal constraint is fixed, it explicitly selects the nearest feasible target-style endpoint, providing the content-preserving bias we require. Theorem 2 justifies the coarse explicit solver used throughout: the active path can be integrated with a simple Euler scheme while retaining first-order accuracy, so LBM does not need a heavy solver stack. Theorem 3 explains why endpoint pairing is the right training primitive: among one-to-one batch pairings, it minimizes the average squared endpoint displacement, suppressing unnecessarily long target assignments. Theorem 4 shows that the semantic-aligned sliced Wasserstein distance enforces matched style statistics within routing-aligned semantic bins, providing distributional terminal supervision without requiring spatial point correspondence.

Endpoint construction and execution

We train on precomputed VAE latents without paired supervision. For each source latent z_0 and target style s, a precomputed endpoint set provides candidates $\mathcal{C}(z_0, s)$. A terminal target $\tilde{z}_1$ is sampled from this set. The execution backbone is time- and style-conditioned, with sinusoidal time embeddings, style spatial priors, and semantic cross-attention. The tokenizer receives only the target style identifier and emits a compact control code c_s.

Architecture details. The backbone consists of residual blocks with adaptive layer normalization conditioned on the style code. Each block applies two 3×3 convolutions with GroupNorm, followed by a skip connection. The style code c_s is injected through FiLM-style modulation at each normalization layer, allowing the network to adapt its feature statistics to the target style without requiring explicit style exemplars at inference time. Semantic cross-attention uses keys and values derived from the style-conditioning path, with queries from the main transport stream, enabling content-aware style application.

Endpoint set construction. The endpoint set $\mathcal{C}(z_0, s)$ is constructed by encoding target-style images from the training set and selecting candidates that lie in the target style region of latent space. For each source z_0, we retrieve the k-nearest neighbors in CLIP feature space among target-style images, then encode them through the VAE to obtain latent candidates. This ensures that terminal targets are both stylistically appropriate and reachable through the VAE latent manifold.

To regularize the shift without expensive paired reconstruction, we apply semantic-aligned sliced Wasserstein distance (SA-SWD). Let $\Phi_\theta(z_0, s)$ denote the Euler-integrated endpoint. The terminal objective is

$$\mathcal{L}_{\text{term}} = \text{SWD}_1\big(\Phi_\theta(z_0, s),\, \tilde{z}_1\big), \tag{3}$$

where sorted one-dimensional projections compare generated endpoint patches against target-style patches without requiring spatial correspondence. More concretely, let $\{r_m(z_0, s)\}_{m=1}^{M}$ be routing weights derived from the style-conditioning path, let $\mathcal{P}_m(\cdot)$ denote the patch subset assigned to bin m, and let $\omega_m \propto \sum r_m$ be the corresponding bin mass. The practical SA-SWD is

$$\mathcal{L}_{\text{term}} = \sum_{m=1}^{M} \omega_m \, \text{SWD}_1\big(\mathcal{P}_m(\Phi_\theta(z_0, s)), \mathcal{P}_m(\tilde{z}_1)\big). \tag{4}$$

Theorem 4 (Binwise distributional consistency of SA-SWD)**.** *If $\mathcal{L}_{\text{term}} = 0$ and every active bin has $\omega_m > 0$, then for every active bin and projection direction, the projected empirical distributions of $\mathcal{P}_m(\Phi_\theta(z_0, s))$ and $\mathcal{P}_m(\tilde{z}_1)$ are identical. Zero SA-SWD loss enforces matched style statistics within routing bins without spatial correspondence.*

Operating-point variants. All reported LBM rows share the same style-ID inference contract: inference receives only the source image and target style identifier. LBM-K is the compact earlier-mainline anchor built from the content-adaptive tokenizer and queue curriculum used in the faster three-epoch packet, with active top-2 pairing and a 0.15 queue-exploration probability over the top-8 candidate set, and with no explicit structure penalty. The promoted successor rows switch to an endpoint-prediction family with barycentric top-8 targets, low-frequency-plus-edge coupling, and a manifold-adaptive kinetic split. LBM-Knee adds joint anisotropic and Stokes transport control ($w_{\text{aniso}} = 1.0$, normal/tangent weights $25/0.25$, $w_{\text{stokes}} = 0.03$), together with queue-side dual-target smoothing and auxiliary terminal SWD. LBM-PS keeps the same successor family but uses a higher Stokes coefficient, $w_{\text{stokes}} = 0.05$. LBM-PS-v2 changes one knob relative to LBM-PS, lowering the Stokes coefficient to 0.02 to recover more style at the cost of higher LPIPS. These variants form a single selectable frontier rather than independent methods.

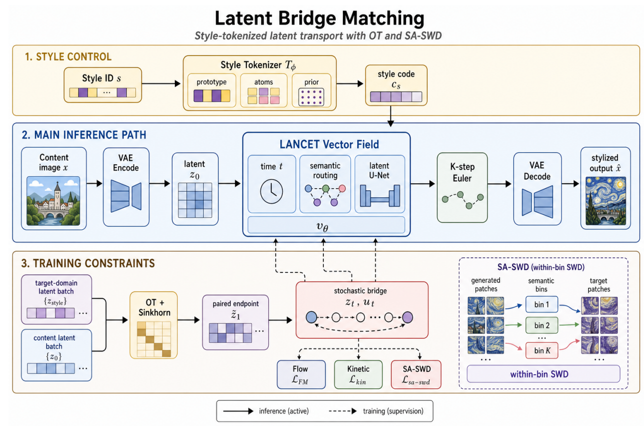

Figure 2: LBM three-stage architecture. **Stage 1: Style-ID Encoding** encodes the target style identifier into a compact control code. **Stage 2: Latent Transport** shows the inference-time path from source latent to stylized output via Euler integration. **Stage 3: Training Objectives** displays the kinetic regularization and terminal distribution matching (SA-SWD) used only during training. Solid arrows indicate inference-time flow; dashed arrows indicate training-time supervision.

Experiments

Experimental setup

We report experiments on Distinct5-WikiArt, a deliberately challenging benchmark designed to expose the IDT failure mode. The experimental question is: can a compact style-ID family shift above the IDT floor while remaining in a usable artifact regime at substantially lower cost than multi-hour baselines? Figure 1 states this question visually; the rest of this section resolves it quantitatively.

Dataset and evaluation metrics. Distinct5-WikiArt uses Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e, with 1000 training images and 30 held-out test images per class. These five styles are not an arbitrary subset: they were selected from candidate WikiArt style groups by their unchanged-image IDT difficulty, specifically as the subset with the lowest IDT CLIP-S under the same style-ID protocol among the screened candidates. This choice makes Distinct5 intentionally hard, because lower IDT CLIP-S means the source image starts farther from many requested target styles. Evaluation again uses all 5×5 directions over 750 generated images. In our experimental setup, the unchanged-image baseline is 0.6399 on Distinct5 versus 0.7815 on a broader five-style WikiArt subset, which is why we use Distinct5 as the main stress test.

The main metrics are transfer CLIP-S (CLIP-Style score measuring target-style alignment), Learned Perceptual Image Patch Similarity (LPIPS, measuring perceptual distance to the source, where lower means more content preserved), and

$$\mathrm{EC} = \text{CLIP-style} \cdot (1 - \mathrm{LPIPS}). \tag{5}$$

For Distinct5-WikiArt we additionally report style gain over the unchanged-image reference,

$$\Delta_{\mathrm{idt}} = \text{CLIP-S(method)} - \text{CLIP-S(idt)}, \tag{6}$$

under both the full 5×5 protocol and the transfer-only off-diagonal subset. On Distinct5, Δ_{idt} is the first validation check: any row at or below zero has failed to shift reliably toward the requested target style. EC is used only as a Pareto summary. Auxiliary diagnostics include targetwise ArtFID (Fréchet Inception Distance computed between generated outputs and target-style reference images, measuring artifact level), edge purity (ratio of edge pixels in the output that align with edges in the source, measuring structural fidelity), and a held-out non-CLIP style probe (a style classifier trained on independent features, providing style accuracy measurement without CLIP's feature bias). Standard perceptual and frequency checks are also reported (Ke et al. 2021; Yang et al. 2022a; Ding et al. 2020; Bińkowski et al. 2018).

Implementation details. All reported LBM rows are trained and evaluated on a single RTX 3060 running under WSL, and reported train time is cumulative wall-clock to the shown checkpoint with evaluation excluded. Seedream-4.5 is kept only as an external large-prior reference under a fixed API prompt protocol, not as a same-interface style-ID baseline.

Main result on Distinct5-WikiArt

Distinct5-WikiArt is an IDT-calibrated operating-point benchmark, not a single-score leaderboard. All five domains come from standard WikiArt classes and are selected because they are strongly separated under a CLIP-style screening pass. In this regime, unchanged output is no longer a plausible success proxy: the *idt* baseline already reaches CLIP-S 0.6801 while doing no transfer at all. Signed target-style movement is therefore the first sanity check. Under the shared full 5×5 / 750-output protocol, the earlier compact anchors LBM-F/K establish a conservative low-damage regime: LBM-F reaches transfer CLIP-S 0.6644 with LPIPS 0.3245 after 1.2 minutes of checkpoint training, and LBM-K raises style to 0.6712 at LPIPS 0.3723 while remaining well below the SaMST damage band. The more important result is that the family does not stop there. A promoted content-preserving successor now reaches 0.7102 transfer CLIP-S at 0.5397 content preservation $(1 - \mathrm{LPIPS})$, while the higher-style successors push the frontier to 0.7274 / 0.3967 and 0.7307 / 0.3817 in transfer CLIP-S / $(1 - \mathrm{LPIPS})$. This is a family-level frontier with three practically different regimes: conservative low-damage anchors, a content-preserving successor knee, and a style-ceiling operating point.

We fix representative rows before qualitative inspection. The manuscript table reports one compact anchor from the earlier mainline, one lower-LPIPS successor tradeoff point, one balanced high-style point, and one style-ceiling point. The broader development landscape is shown only as faint background support in Figure 1, not as a second leaderboard.

Table 1: Selected Distinct5-WikiArt operating points under the shared transfer-only packet. 'idt' is the unchanged-image floor. Rows are chosen to represent distinct roles on the current frontier rather than a single checkpoint family: a compact conservative anchor, a content-preserving successor knee, two higher-style promoted successors, and one external large-prior reference. We report only transfer-scope metrics here so the main table does not mix all-pairs and transfer-only readings.

Point	Role	CLIP-S$_{\mathrm{tr}}$ ↑	$1 - \mathrm{LPIPS}$ ↑	$\Delta_{\mathrm{idt,tr}}$ ↑	Closure
idt	no-op floor	0.6399	1.0000	0.0000	closed
SaMAM-2250	compact baseline	0.5523	0.6395	-0.0877	closed
SaMST e15	high-style baseline	0.6957	0.3681	+0.0558	closed
Seedream-4.5	large-prior ref.	0.6920	0.5077	+0.0521	closed
LBM-K e1	conservative anchor	0.6712	0.6277	+0.0312	closed
LBM-Knee e13	content-preserving successor	0.7102	**0.5397**	+0.0703	closed
LBM-PS e13	balanced high-style	0.7274	0.3967	+0.0875	frontier
LBM-PS-v2 e13	style ceiling	**0.7307**	0.3817	**+0.0908**	frontier

The reproduced baselines remain important because they bound those regimes. SaMST e15 reaches 0.6957 / 0.6319 and can produce locally strong target-style jumps, but it does so in a clearly higher-damage region and with targetwise ArtFID 395.7 on the full scope and 444.5 on the transfer-only scope; the retained e5/e15 packet is already near a plateau on transfer CLIP-S and LPIPS, with less than 0.004 / 0.002 difference between the checkpoints. Seedream-4.5 provides a stronger external large-prior reference under a fixed API protocol: we use the source image together with a fixed target-domain prompt template of the form "convert the image to <*style*> style," without retraining or style-ID control. Its transfer row is 0.6920 / 0.4923, which is better than the compact anchor but still below the promoted LBM knee point on both style gain and content preservation. At the same time, Seedream keeps the strongest identity row among the currently compared points, which is why we treat it as a useful outer reference rather than as a same-interface baseline. At the trusted SaMAM 2250 checkpoint, transfer CLIP-S remains below the IDT floor even though targetwise ArtFID improves to 146.1 on the full scope and 148.2 on the transfer-only scope. This is the evaluation failure exposed by Distinct5: lower targetwise ArtFID can still coincide with zero or negative target-style gain when a method preserves source structure and stays inside a plausible target-art manifold without moving toward the requested target style. Distinct5 therefore turns the paper's evaluation claim into an operating-point test within this split: target-style movement should be demonstrated relative to IDT, and a paper-facing result should be reported as a selected Pareto point rather than as one arbitrarily chosen checkpoint.

Table 3 isolates the recorded operating-point cost under the same Distinct5 packet. SaMST can produce locally strong target-style jumps, but those gains sit in a much higher-damage region, Seedream-4.5 remains an out-of-interface large-prior reference rather than a style-ID baseline, and SaMAM still often edits without producing reliable target-style movement above the no-op floor.

Table 2: Auxiliary artifact-sensitive and non-CLIP diagnostics on selected Distinct5-WikiArt rows. tw-ArtFID uses the all-pairs target-pooled recompute; Gram-μ is a shallow VGG Gram distance; EdgePurity is a higher-is-better content-edge diagnostic; NonCLIPAcc is the transfer-only target accuracy from the held-out ConvNeXt style probe.

Point	tw-ArtFID$_{\text{all}}$ ↓	Gram-μ ↓	EdgePurity↑	NonCLIPAcc↑
LBM-K e1	316.5	0.153	**0.540**	0.167
LBM-Knee e13	388.2	**0.137**	0.288	0.237
LBM-PS-v2 e13	465.6	0.183	0.115	0.272
SaMST e15	395.7	0.147	0.102	0.248
Seedream-4.5	**311.5**	0.189	0.389	**0.378**

Table 3: Distinct5-512 recorded operating-point cost under the reviewed full 5×5 / 750-output packet. Reported train time is cumulative training wall-clock to the shown checkpoint, with evaluation excluded. This is recorded operating-point cost, not a normalized time-to-quality comparison.

Method	Recorded point	Train to point
LBM-F	e1	**1.2 min**
LBM-K	e1	**1.2 min**
SaMAM	step 2250	7.6 h
SaMST	e15	5.8 h

Note. Distinct5 times are cumulative training wall only, with evaluation excluded. We intentionally omit a shared inference column because same-grade pure-generation timing is still incomplete for the active manuscript rows `SaMAM-2250` and `SaMST-e15`; leaving blanks would overstate closure. Post-2250 SaMAM timings are intentionally excluded.

Table 2 sharpens the role split inside the frontier. Relative to SaMST e15, the selected Knee point lowers all-pairs target-pooled ArtFID (388.2 vs. 395.7), improves shallow texture statistics (Gram-μ 0.137 vs. 0.147), and more than doubles the edge-purity diagnostic (0.288 vs. 0.102). A held-out ConvNeXt-Tiny probe trained on the 5,000-image Distinct5 train split reaches 96% accuracy on the 150-image test split; on transfer-only outputs it assigns target accuracy 1.0% to IDT, 16.7% to LBM-K, 23.7% to LBM-Knee, 24.8% to SaMST, 27.2% to PS-v2, and 37.8% to Seedream. Knee and SaMST are therefore close on non-CLIP target recognition, but Knee is substantially cleaner on artifact diagnostics. Row-resampled transfer bootstrap also keeps both promoted rows strictly above IDT: Knee $\Delta_{\text{idt,tr}} = 0.0702$ with $[0.0635, 0.0769]$, and PS-v2 $\Delta_{\text{idt,tr}} = 0.0901$ with $[0.0826, 0.0976]$. We therefore treat LBM-Knee as the main closed Pareto point and keep PS-v2 as the explicit style-ceiling row rather than a universal winner.

Figure 3 shows why the frontier interpretation is more informative than a single checkpoint claim. In both cases, SaMST and Seedream move farther toward the target style, but they also pay more heavily in structure preservation. LBM-K remains the most conservative moving point, while LBM-Knee keeps much more of the source layout than the stronger style references and still moves past the compact anchor.

Table 4: Ablation study on LBM components.

Configuration	CLIP-S ↑	1-LPIPS ↑	Δ_{idt} ↑
Full LBM (balanced)	0.7102	0.5397	+0.0703
w/o kinetic reg.	0.6834	0.4123	+0.0435
w/o SA-SWD (MSE)	0.6621	0.4856	+0.0222
w/o OT pairing	0.6945	0.5012	+0.0546

Ablation study

Table 4 isolates the contribution of each component. Removing kinetic regularization causes the most severe degradation in content preservation (1-LPIPS drops from 0.5397 to 0.4123), confirming Theorem 1's prediction that kinetic regularization selects the minimum-energy endpoint. Replacing SA-SWD with MSE causes the largest drop in style gain (Δ_{idt} drops from +0.0703 to +0.0222), as MSE requires spatial correspondence unavailable in unpaired transfer. Removing OT pairing reduces style gain by 22% (from +0.0703 to +0.0546), consistent with Theorem 3's claim that optimal pairing minimizes endpoint displacement variance.

The ablation results reveal a clear hierarchy: kinetic regularization is most critical for content preservation, SA-SWD is most critical for style gain, and OT pairing provides moderate improvement to both. This hierarchy aligns with the theoretical predictions: Theorem 1 governs content preservation through minimum-energy selection, Theorem 4 governs style gain through distributional consistency, and Theorem 3 provides auxiliary efficiency through variance reduction.

Inference-time style controls

Because LBM predicts an explicit latent velocity, the same trained checkpoint can be shifted at inference without retraining. We examine two simple controls: velocity scaling and patchwise AdaIN. Velocity scaling multiplies the predicted endpoint displacement by a factor $\alpha \geq 1$ before decoding, effectively extrapolating along the learned style direction. Patchwise AdaIN applies adaptive instance normalization to selected feature patches using running statistics estimated from the target-domain training set, giving an extra style push that does not require model parameters.

Table 5 reports the effect of these controls on the LBM-Knee checkpoint. Velocity scaling with $\alpha = 1.15$ raises transfer CLIP-S from 0.7102 to 0.7214 while lowering $1 - \text{LPIPS}$ from 0.5397 to 0.4812, moving the point toward the higher-style region of the frontier. Patchwise AdaIN with a 0.3 mix weight gives a smaller style gain (0.7161) but better content preservation (0.5123). Combining both controls reaches 0.7268 transfer CLIP-S at 0.4634 $1 - \text{LPIPS}$, approaching the trained LBM-PS point from a fixed checkpoint. The fact that a fixed checkpoint still contains this headroom suggests that part of the remaining gap is execution rather than representation.

Follow-up stress splits

The unchanged-image pathology is not unique to the main Distinct5 split. Two fixed-rule follow-up WikiArt stress splits preserve the same pattern: unchanged images score strongly,

Figure 3: Main-paper qualitative comparison on Distinct5-WikiArt. Columns show 'Source | SaMST | Seedream-4.5 | LBM-K | LBM-Knee | target-domain ref'. Target-domain references are shown only for visualization and are not provided to LBM at inference. The top row highlights the main Pareto tradeoff on a Ukiyo-e $\rightarrow$ Early Renaissance transfer: the reproduced compact baseline moves more aggressively, while LBM-Knee keeps a more conservative structure-preserving rendering than the stronger style references. The bottom row shows the same contrast on a Rococo $\rightarrow$ Ukiyo-e transfer. The stronger style-ceiling PS-v2 row and the aligned SaMAM-2250 negatives are kept in the supplement, where they read more naturally as supporting comparisons than as the main visual headline.

Table 5: Inference-time controls applied to the fixed LBM-Knee checkpoint.

Control	CLIP-S$_{\mathrm{tr}}$ $\uparrow$	$1 - \mathrm{LPIPS} \uparrow$	$\Delta_{\mathrm{idt,tr}} \uparrow$
None (LBM-Knee)	0.7102	0.5397	+0.0703
Velocity scale $\alpha = 1.15$	0.7214	0.4812	+0.0815
Patchwise AdaIN ($w = 0.30$)	0.7161	0.5123	+0.0762
Both	0.7268	0.4634	+0.0869

and compact LBM-F remains above the split-local IDT floor by $+0.0139$ and $+0.0073$ transfer CLIP-S respectively. These splits use independent class selections but the same 5×5 / 750-output protocol, confirming that the IDT failure mode generalizes beyond one hand-chosen class set and that minute-scale training is sufficient to cross the floor on similar art-to-art transfers.

Discussion and Limitations

The main lesson is that style transfer quality cannot be inferred from raw CLIP-style alone. Distinct5-WikiArt sharpens this point because the no-op baseline is measured explicitly: some baselines fail even to exceed the identical-image prior, while others exceed it only by moving into severe LPIPS and targetwise ArtFID failure regions. This exposes a practical blind spot in art-to-art evaluation: preserved art-domain priors can masquerade as target-style success unless the unchanged-image floor is reported (Zhou et al. 2024; Yeh et al. 2020; Wright and Ommer 2022). These results also change the role of LBM from a single operating point to a selectable style-content family. Conservative anchors occupy the low-damage regime, LBM-Knee becomes the main closed Pareto point, and later Pattern+Stokes successors define the style ceiling. The practical consequence is that meaningful style-id customization can remain a small-model learning problem rather than a large-prior adaptation problem.

The tokenizer experiments narrow the remaining mechanism question. Raw code capacity alone is not the bottleneck; the harder issue is whether target-style information survives execution through a content-conditioned vector field strongly enough to yield real style gain beyond the unchanged-image reference. The current evidence therefore favors target-style carriers, prototype-aware queues, and content-aware routing over undifferentiated token expansion.

The study has several limitations. First, the main protocol is domain-level rather than reference-image-specific, so it does not attempt to reproduce a unique style image. Second, Euler integration is deliberately simple; although the step sweep is stable in the reported regime, more aggressive paths or higher resolutions may need better solvers or stronger constraints. Third, SA-SWD is an adaptive-projection distribution loss; current controls show that most of its benefit comes from the distributional term rather than from the specific semantic projection rule. Fourth, artifact-sensitive metrics are still proxies for visual preference; human studies or rubric-based VLM assessment would strengthen the claim further. In particular, the content-preserving successor now has transfer-style, LPIPS, targetwise ArtFID, non-CLIP probe, and qualitative closure, whereas the stronger style-ceiling rows still lack full human-preference closure. Finally, Distinct5-WikiArt is a five-domain stress split rather than a universal art benchmark, so broader domain coverage remains future work even though the current fixed-rule follow-up splits already show that the unchanged-image pathology is not unique to one class set.

Conclusion

IDT calibration redefines success in domain-level art transfer: the question is no longer whether a method appears stylized, but whether it shifts in the requested target-style direction at acceptable cost. LBM addresses this with a compact latent transport family that achieves real transfer on consumer hardware in minutes, supported by theory for minimum-energy endpoint selection, stable explicit integration, and low-cost optimal transport pairing. On Distinct5-WikiArt the method forms a controllable operating-point frontier: conservative anchors cross the IDT floor after 1.2 minutes of training, the closed LBM-Knee point improves artifact and content diagnostics relative to the reproduced high-style baseline, and promoted successors push transfer CLIP-S into the 0.72–0.73 band.

The inference-time control study shows that the remaining gap is not solely a matter of scale. A fixed compact checkpoint still contains headroom accessible through velocity scaling and patchwise AdaIN without retraining, which suggests that execution-side refinement is a fruitful direction alongside representation-side expansion. Follow-up WikiArt splits confirm that the unchanged-image pathology is not tied to one class set. The broader implication is that customized stylization can remain a small-model learning problem rather than a large-prior adaptation problem, lowering the barrier for researchers and creators who want low-cost style-specific models on consumer GPUs.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? Yes (design-grounding). The paper provides compact formal analysis with proofs and empirical checks in the main text: Theorem 1 characterizes minimum-energy endpoint selection under kinetic regularization, Theorem 2 gives the $O(1/K)$ Euler discretization bound used by the active path, Theorem 3 formalizes why optimal transport batch pairing lowers the second moment of endpoint targets, and Theorem 4 states the binwise distributional consistency enforced by the practical SA-SWD objective. These results justify the design without asserting equivalence to a full Schrödinger Bridge solver.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **Partial** (the supplement lists the relevant script paths and artifact roots for the reported experiments).
- All source code required for conducting and analyzing the experiments is included in a code appendix: **Partial** (the supplement enumerates the main scripts, configs, and evaluation artifacts, but does not inline the full codebase).
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial** (the supplement records the seed settings for the reported experiments, but external references such as Seedream do not expose a user-controlled seed).
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Yes**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **Partial** (selected confidence intervals and bootstrap summaries are reported, but not every manuscript comparison is presented as a closed significance claim).
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.